MILITARY SPECIFICATION

INITIATOR, MECHANICAL TIME FUZE XM109 (10 SEC. TIMER)

1. SCOPE

1.1 This specification covers Initiator, Mechanical Time Fuze XM109 (10 Sec. Timer)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-331	- Fuze and Fuze Components, Environmental and Performance Test for
MIL-STD-810	- Environmental Test Methods
MIL-STD-1167	- Ammunition Data Card

DRAWINGS

Munitions Command

FD 23771	- Initiator, Mechanical Time Fuze XM109
IEL 23771	- Index of Inspection Equipment List for Initiator, Mechanical Time Fuze XM109

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

FSC 1390

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2.2 Other Publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface Texture, Surface Roughness, Waviness and Lay

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies.

3. REQUIREMENTS

3.1 General.-Material, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced standards, and specifications, except as provided in 6.1.

3.2 Safety.-The initiator shall be incapable of functioning, with the pull wire and roll pin in place at any time setting from one (1) to ten (10) seconds.

3.3 Function.-The initiator shall function satisfactorily as determined by timing accuracy at any time setting from one (1) to ten (10) seconds in the temperature range of minus -40°F. to plus +160°F.

3.3.1 Timing Accuracy.-The arming pin in the initiator shall be released at any time setting between one (1) to ten (10) seconds within the accuracy of plus or minus ±0.5 seconds.

3.4 Environmental

3.4.1 Jolt.-The initiator shall comply with MIL-STD-331 Test No. 101, with the exception; the initiator shall be tested as Head Assembly Drawing FF12708.

3.4.2 Jumble.-The initiator shall comply with MIL-STD-331 Test No. 102, with the exception; the initiator shall be tested as a Head Assembly Drawing FF12708 and mounted on a suitable testing base.

3.4.3 Forty-foot Drop.-The initiator shall comply with MIL-STD-331 Test No. 103 with the exception; the initiator shall be inserted into a simulated usage vehicle and dropped in the horizontal position only.

3.4.4 Transportation-Vibration.-The initiator shall comply with MIL-STD-331 Test No. 104, Procedure II.

3.4.5 Humidity.-The initiator shall comply with MIL-STD-810 Method No. 507 with the exception; the initiator shall be mounted on a suitable test plate with a "O" ring seal.

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3.4.6 Rain.-The initiator shall comply with MIL-STD-810 Method No. 506 with the exception; the initiator shall be mounted on a suitable test plate with a "O" ring seal.

3.4.7 Five-foot Drop.-The initiator shall comply with MIL-STD-331 Test No. 111, with the exception; the initiator shall be inserted into a simulated usage vehicle and dropped in the horizontal position only.

3.4.8 Aircraft Vibration.-The initiator shall comply with MIL-STD-810 Method No. 514 Curve C.

3.4.9 Low Temperature.-The initiator must be safe and operable when subjected to a temperature of minus -40°F. -5°F. for twenty-four (24) hours.

3.4.10 High Temperature.-The initiator must be safe and operable when subjected to a temperature of plus +160°F +5°F. for twenty-four (24) hours.

3.5 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specification and the following:

3.5.1 Metal Defects.-All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defects.

3.5.2 Surface Finish.-The requirements for surface finish are as detailed on the applicable drawings and ASA-B46.1.

3.5.3 Burrs.-No parts shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.4 Foreign Matter.-No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter. The cleaning methods and agents shall not adversely affect the finish nor contaminate the parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Quality Assurance Terms and Definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 First Article Sample

4.2.1 Initial Production Sample.-At the beginning or regular production, when required by the contract, a sample consisting of the following shall be submitted:

100 component parts as delineated in 4.2.2
275 subassemblies as delineated in 4.2.3
40 initiators as delineated in 4.2.5

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and material, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and Test.-After inspection by the contractor and provisional acceptance at source by the Government, one hundred (100) component parts, two hundred and seventy-five (275) subassemblies and forty (40) initiators shall be inspected, as provided herein, at a Government laboratory or such other facility as specified in the contract. The government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.2.2 Components

4.2.2.1 Twenty-five (25) of each component shall be tested: 4.3.3.1 (Hardness), 4.3.3.2 (Load) and 4.3.3.4 (Torque).

4.2.3 Subassemblies

4.2.3.1 Twenty-five (25) of each subassembly shall be tested as in 4.3.3.3 (Static Load), 4.3.3.5 (Pull) and 4.3.3.6 (Push).

4.2.4 Acceptance Criteria.-Failure of any component or subassembly to comply with the requirements of the applicable drawing for the above characteristics shall be cause for rejection of the initial production sample.

4.2.5 Initiators

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4.2.5.1 Timing Accuracy.-Each initiator, prior to the following tests, shall be set at ten (10) seconds and subjected to the timing accuracy test. Failure of the initiator to comply with the timing accuracy requirement (3.3.1) shall be cause for the rejection of the unit.

4.2.5.2 Environmental Tests.-Forty (40) initiators shall be set at five seconds and then subjected to the following MIL-STD tests.

Test	Quantity	Requirement	Test Method
Jolt	4	3.4.1	4.4.8.1
Jumble	4	3.4.2	4.4.8 2
Forty-Foot Drop	4	3.4.3	4.4.8.3
Five-Foot Drop	4	3.4.7	4.4.8.9
Transportation-Vibration	4	3.4.4	4.4.8.8
Aircraft Vibration	4	3.4.8	4.4.8.7
Low Temperature	4	3.4.9	4.4.8.4
High Temperature	4	3.4.10	4.4.8.4
Humidity	4	3.4.5	4.4.8.5
Rain	4	3.4.6	4.4.8.6

4.2.5.3 Acceptance Criteria.-Occurance of any of the following in any initiator shall be cause for rejection of the initial production sample.

4.2.5.3.1 Jolt, Jumble and Forty-foot Drop.-

a. Arming pin assembly functioned or disassembled
b. Stop pin missing (roll pin)
c. Pull wire broken or damaged
d. Any broken, deformed or displaced parts which renders the initiator unsafe to handle or dangerous to use.

4.2.5.3.2 Five-foot Drop.-Transportation-Vibration, Aircraft Vibration, Low Temperature and High Temperature.-

a. Arming pin assembly functioned or disassembled.
b. Initiator moved from the five (5) second set position.
c. Failure of any assembly to comply with the timing accuracy requirements (3.3.1) after the tests.
d. Pull wire broken, damaged or missing.
e. Broken, deformed or displaced parts which renders the initiator unsafe to handle or dangerous to use.

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4.2.5.3.3 Humidity and Rain.-

a. Conditions listed under 4.2.5.3.2
b. Any excessive moisture; discoloration on any plated parts, fungi, etc.

4.3 Inspection Provisions

4.3.1 Submission of Product.-The product shall be presented in accordance with MIL-STD-105.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with MIL-STD-105. For the classification of defects contained herein the A.Q.L. for each shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis for surface roughness determination.

4.3.2.1 Head Drawing FD-23740 covering a detail of Drawing FD-23771.

	Defect	Method of Inspection
MAJOR	A.Q.L. = .40%	
101.	Eccentricity of movement counterbore with datum "A"	Gage
102.	Diameter of arming pin hole less than min.	Gage
103.	Eccentricity of base locating boss with datum "A"	Gage
104.	Diameter of locating boss exceeds max.	Gage
105.	Position of mounting holes beyond basic dimensions	Gage
106.	Diameter of mounting holes below min.	Gage
107.	Diameter of movement cavity less than min.	Gage
108.	Metal burred, damaged or defective	Visual
109.	Foreign matter	Visual

4.3.2.2 Arming Pin Drawing FD12944 covering a detail of Drawing FD23771.

MAJOR	A.Q.L. = .40%	
101.	Overall length of arming pin less than min.	Gage
102.	Body diameter of arming pin exceeds max.	Gage
103.	Head diameter of arming pin exceeds max.	Gage
104.	Metal burred, damaged or defective	Visual

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4.3.2.3 Balance and Pin Assembly, Drawing FC12945 covering a detail of Drawing FD23771.

	Defect	Method of Inspection
MAJOR	AQL = .40%	
101.	Extension of balance stop pins less than min.	Gage
102.	Length of balance pins exceed max.	Gage
103.	Overall length of balance and balance shaft exceeds max.	Gage
104.	Balance pins to be parallel with datum "A"	Gage
105.	Crimping of balance weight inadequate	Visual
106.	Insecure, improper or incomplete assembly	Visual

4.3.2.4 Movement Assembly Drawing FF12709, covering a detail of Drawing FD23771.

MAJOR	AQL = .40%	
101.	Distance from base to shoulder max.	Gage
102.	Housing diameter exceeds max.	Gage
103.	Housing to be concentric with datum "A" and perpendicular to datum "B"	Gage
104.	Insecure, improper or incomplete assembly	Manual/Visual
105.	Metal burred, damaged, defective	Visual
106.	Foreign matter	Visual

4.3.2.5 Head Assembly FF12708 covering a detail of Drawing FD23771.

MAJOR	AQL = .40%	
101.	Roll pin (safety) missing or insecure	Visual
102.	Improper spinning of head to dial plate	Manual/Visual
103.	Metal burred, damaged or defective	Visual
104.	Dial marking incorrect or illegible	Visual

4.3.2.6 Initiator, Drawing FD23771

MAJOR	AQL = .40%	
101.	Pull wire insecure; loops missing	Visual
102.	Improper assembly of cap and pull wire	Visual
103.	Metal or plastic burred, damaged or defective	Visual
104.	Sealant missing, loose or improperly applied	Visual

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4.3.2.7 Process Control Audit.-Periodic unscheduled monitoring shall be conducted for the process operations on the following assemblies to determine compliance with the requirements of the applicable drawings.

a. Mainspring FC12950

(1) Mainspring torqueing operation

b. Movement Assembly FF12709

(1) Application of lubricant
(2) Assembly of mainspring retainer to housing assembly
(3) Pre-winding of the mainspring
(4) Insertion of release plunger and spring
(5) Insertion of the pull wire

c. Head Assembly FF12708

(1) Assembly of head, movement and dial plate
(2) Inspection of zero setting
(3) Insertion of arming pin and spring
(4) Rotation of dial indicator
(5) Insertion of (safety) roll pin

d. Initiator FD23771

(1) Assembly of the cap
(2) Assembly of the pull wire with loops at both ends
(3) Application of sealant

4.3.3 Inspection Testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Hardness (See 4.4.1 - Destructive).-Twenty-five (25), spring washers from each lot of components shall be subjected to the hardness specified on drawing FB55723. Failure of any component to comply with the requirement shall be cause for rejection of the lot of washers.

4.3.3.2 Load (See 4.4.2 - Destructive).-Twenty-five (25) samples of the following components from each lot shall be subjected to the load test. Failure of any component to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot of components.

Spring, release plunger	FB55685
Spring, arming pin	FB55721

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4.3.3.3 Static Load Tests (See 4.4.3 - Destructive).-Twenty-five (25) samples of the following components shall be subjected to the load test. Failure of any component to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot of components.

Pull wire assembly	FB55724
Escape wheel and pinion assembly	FC12952
Gear and pinion No. 1 assembly	FC12954
Gear and pinion No. 2 assembly	FC12963
Gear and pinion No. 3 assembly	FC12989
Gear and pinion No. 4 assembly	FC12991
Top plate, pin and gear assembly	FD23744
Housing assembly	FD23743
Movement assembly	FF12709

4.3.3.4 Torque Tests (See 4.4.4).-Twenty-five (25) mainsprings shall be subjected to the torque test specified on drawing FC12950. Failure of any component to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot of mainsprings.

4.3.3.5 Pull Test (See 4.4.5 - Destructive).-Twenty-five (25) pin balance stops and pin balance of the Balance and Pin Assembly shall be subjected to the pull test specified on drawing FC12945. Failure of a component to comply with the requirement shall be cause for rejection of the lot of balance and pin assemblies.

4.3.3.6 Push Test (See 4.4.6 - Destructive).-Twenty-five (25) guide arming pins, of the arming pin assembly, shall be subjected to the push test specified on drawing FC12944. Failure of any assembly to comply with the requirement shall be cause for rejection of the lot of arming pin assemblies.

4.3.3.7 Timing Accuracy (See 4.4.7).-Each initiator shall be set at ten (10) seconds and subjected to the timing accuracy test. Failure of the initiator to comply with the timing accuracy requirement (See 3.3.1) shall be cause for rejection of the assembly.

4.3.3.8 Environmental Tests (See 4.4.8 - Destructive).-Forty (40) initiators from each production lot shall be set at five (5) seconds and then subjected to the following MIL-STD-tests:

Test	Quantity	Requirement	Test Method
Jolt	4	3.4.1	4.4.8.1
Jumble	4	3.4.2	4.4.8.2
Forty-foot Drop	4	3.4.3	4.4.8.3
Five-foot Drop	4	3.4.7	4.4.8.9
Transportation-Vibration	4	3.4.4	4.4.8.8

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4.3.3.8 (Cont'd)

Test	Quantity	Requirement	Test Method
Aircraft Vibration	4	3.4.8	4.4.8.7
Low Temperature	4	3.4.9	4.4.8.4
High Temperature	4	3.4.10	4.4.8.4
Humidity	4	3.4.5	4.4.8.5
Rain	4	3.4.6	4.4.8.6

4.3.3.8.1 Acceptance Criteria.-Production Lot acceptance criteria shall be as specified in paragraph 4.2.5.3.

4.3.3.9 Inspection Equipment.-The examination and tests shall be made using equipment listed in IEL23771. Equipment design shall be in accordance with code designation on the applicable IEL.

4.4 Test Method and Procedures

4.4.1 Hardness.-Samples shall be tested in an area which is solidly supported in order to obtain valid readings. The median value of three hardness readings taken in each sample unit shall be used as the individual value.

4.4.2 Load.-The test shall be performed using the values as specified on drawing FB55685 and FB55721. The springs that have been subjected to the load test shall be scrapped.

4.4.3 Static Load.-The test shall be performed using the values as specified on the applicable drawings. All samples tested shall be scrapped.

4.4.4 Torque.-The mainspring shall be assembled to the retainer mainspring drawing FC12997 and subjected to the required torque value.

4.4.5 Pull Test.-The pin balance stop shall be tested using the values as specified on drawing FC12945. The balance assemblies that have been subjected to the pull test shall be scrapped.

4.4.6 Push Test.-The guide arming pin shall be tested using the values as specified on drawing FC12944. The arming pin assemblies that have been subjected to the push test shall be scrapped.

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4.4.7 Timing Accuracy.-The initiator shall be secured to the fixture. The time shall be set to the specified setting. The release of the firing arm shall be recorded using measuring equipment consisting of an electric clock calibrated to .01 second and triggered by suitable micro-switches.

4.4.8 Environmental Test Procedures.-The component parts of the initiator, which have been disassembled and inspected following tests, shall be scrapped.

4.4.8.1 Jolt.-The test shall be performed in accordance with procedures specified in MIL-STD-331 Test No. 101 except as provided in 3.4.1. After the test the initiator shall be inspected as follows:

a. The unit shall be tested for the timing accuracy requirement (3.3.1) and the results recorded for information only.

b. Remove the movement assembly from the head; check for any loose or dislocated parts.

c. Remove mainspring housing; check gear - train for loose subassemblies or broken parts.

4.4.8.2 Jumble.-The test shall be performed in accordance with procedures specified in MIL-STD-331 Test No. 102 except as provided in 3.4.2. After the test the initiator shall be inspected as outlined in paragraph 4.4.8.1.

4.4.8.3 Forty-Foot Drop.-The test shall be perfromed in accordance with procedures specified in MIL-STD-331 Test No. 103 except as provided in 3.4.3. After the test the initiator shall be inspected as follows:

a. Remove the pull wire and plastic cap; check arming pin assembly.

b. The initiator shall be tested for the timing accuracy requirement (3.3.1) and the results recorded for information only.

c. Remove the movement assembly from the head; check for any loose or dislocated parts.

d. Remove mainspring housing; check gear-train for loose subassemblies or broken parts.

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4.4.8.4 Extreme Temperature Phase.-The sample initiators shall be saturated as outlined in paragraph 4.4.8.4.1. The timing accuracy test (3.3.1) shall be performed to meet the requirements of 3.4.9 or 3.4.10 as applicable.

4.4.8 4.1 Temperature Conditioning (-40°F. and +160°F.).-The test chamber shall be at ambient temperature at the start of the test and the chamber maintained at a temperature of minus -40°F. or plus +160°F. as applicable, until stabilization of the initiator temperature is reached (see 6.2). After stabilization the initiator shall be saturated at minus -40°F. or plus +160°F. as applicable, for one hour, minimum.

4.4.8.5 Humidity.-The test shall be performed in accordance with the procedures specified in MIL-STD-810 Method 507 except as provided in 3.4.5. Upon completion of the test the initiator shall be tested for timing accuracy as specified in 3.3.1 and then disassembled for examination of all visible surfaces for moisture, discoloration or fungi.

4.4.8.6 Rain.-The test shall be performed in accordance with the procedures specified in MIL-STD-810 Method 506 except as provided in 3.4.6. After the test the initiator shall be inspected as outlined in paragraph 4.4.8.5.

4.4.8.7 Aircraft Vibration.-The test shall be performed in accordance with the procedures specified in MIL-STD-810 Method No. 514 Curve C. Upon completion of the test, the initiator shall be inspected as follows:

a. Remove the pull wire and plastic cap; check arming pin assembly..

b. Test the initiator for timing accuracy requirement (3.3.1).

c. Remove the movement assembly from the head; check for any loose or dislocated parts.

d. Remove mainspring housing; check gear-train for loose subassemblies or broken parts.

4.4.8.8 Transportation-Vibration.-The initiator shall be tested in accordance with procedures specified in MIL-STD-331 Test No. 104 Procedure II. Upon completion of the test, the initiator shall be inspected as outlined in paragraph 4.4.8.7.

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4.4.8.9 Five-Foot Drop.-The test shall be performed in accordance with the procedures specified in MIL-STD-331 Test No. 111 except as provided in 3.4.7. Upon completion of the test, the initiator shall be inspected as outlined in paragraph 4.4.8.7.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.

5.2 Packing.-To be specified by the contracting agency.

5.3 Marking.-As specified by the contracting agency.

5.4 Data Cards.-Data card information shall be as specified in MIL-STD-1167.

6. DEFINITIONS AND NOTES

6.1 Components manufactured in strict accordance with applicable drawings represent one set of conditions which will result in satisfactory performance. Dimensions which are specifically required by Classification of Defects, as listed in Section 4 and those which affect external profile of assembly, are mandatory. Pertaining to other dimensions, it is recognized that current mass production methods may occasionally produce parts which are not in full compliance with all dimensional requirements but which may provide adequate fuze performance. Therefore, the contractor, at his own risk, may utilize these parts provided there is no adverse affect on timer functionability. This relaxation of requirements applies to "after the fact" deviations only, to dimensions only, on applicable drawings, and does not constitute authority to establish new dimensions or to relax efforts, to achieve the specified limits. The procedure for accomplishing dimensional changes is available under the established engineering change system. Nothing above is to be construed as relieving the contractor of any responsibility in connection with performance of the fuze.

6.1.1 The contractor shall maintain records showing deviations from dimensional requirements, the extent of such deviations, the amount of material involved, and the date the material is introduced into the assembly line. Copies of these records shall be forwarded to the agency assigned custody of the technical data.

6.2 Temperature stabilization of fuzes may be checked by a thermal measuring device in good thermal contact with the largest centrally located internal mass.

6.3 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1390 - A624

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